COMSATS University Islamabad Lahore Campus

Mid-Term Examination – Spring 2025

Course Title:	Database Systems				Course Code:	CSC270	Credit 3+1
Course	Abdul Qayyum				Programme	BSCS	
Semester:	4th	Batch:	FA23	Section:	C, D	Date:	17-04-25
Time Allowed:	90 Minutes				Maximum Marks:		25
Student's					Reg. No.		

Question 1:

{Blooms Taxonomy Learning Level --Understanding *[CLO-1]* [3+2=5 Marks]

a. Compare the roles of logical database designer and physical database designer.
b. Defend why data independence is important in database approach.

Question 2:

{Blooms Taxonomy Learning Level --Applying *[CLO-2]* [3+2=5 Marks]

Left Table

Date	CountryID	Units
1/1/2024	1	40
1/2/2024	1	25
1/3/2024	3	30
1/4/2024	2	35

Right Table

ID	Country
1	USA
2	Canada
3	Panama
4	Spain

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Owner of SuperCUI

a. Apply full outer join on the above two tables and create the resultant table.
b. Create an algebraic form for the following question, use only Left Table to answer. "Display the total number of units against each CountryID"

PTO

Question 3:

{Blooms Taxonomy Learning Level --Applying [CLO-3] **[15 Marks]**

Construct an ERD from the following scenario:

"The owners of "Delicious Pizza shop" want to transfer their business activities on the database systems to get rid of old system. In the old system they were facing delays in delivery, overstaffing, communication gaps and difficulties in order handling. They have different branches in the city, and they want to control them from a single design. First, they want to have data on each branch in the system, each employee's personal and professional details. Second, they require all the pizza details. Third, they should have details of all customers, fourth, they want the record of each sale transaction. A customer can order one or more pizzas in a single transaction. Pizza can be served in the shop or can be delivered to the customer's place/home. The pizza delivery process should also be recorded with the rider's information.

After the implementation of the database system, the owners can handle all the data through their system and can see the different reports like:

- Information about their employes
- Salary of each employee and total monthly salary of each branch
- Which pizza was sold in largest number in any day, week, month or year
- How many pizzas were sold in a day, week, month or year.
- The topmost revenue generated branch per day, week, month or year.
- Etc.

Note: consider and attach relevant and meaningful attributes to each entity.

Construct/Draw an ER diagram to represent the data requirements as following:

- Identify the main entity types. [03]
- Identify the main relationship types between the entity types. [03]
- Identify attributes and associate them with entity or relationship types. [03]
- Determine candidate and primary key attributes for each entity type. [03]
- Identify the occurrence (1 - 1, 1 - *, * - *) between entities. [03]
- State any assumptions necessary to support your design.

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